attractions or repulsions among individuals or where key physical or chemical factors are relatively constant across the study area. Plants established by windblown seeds, such as dandelions, may be randomly distributed in a fairly uniform habitat **(Figure 53.3c)**.

Demographics

The biotic and abiotic factors that influence population density and dispersion patterns also influence other characteristics of populations, including birth, death, and migration rates. **Demography** is the study of these key characteristics of populations and how they change over time. A useful way to summarize demographic information for a population is to make a life table.

Life Tables

A **life table** summarizes the survival and reproductive rates of individuals in specific age-groups within a population. To construct a life table, researchers often follow the fate of a **cohort**, a group of individuals of the same age, from birth until all of the individuals are dead. Building the life table requires determining the proportion of the cohort that survives from one age-group to the next. It is also necessary to keep track of the number of offspring produced by females in each age-group.

Demographers who study sexually reproducing species often ignore the males and concentrate on the females in a population because only females produce offspring. Using this approach, a population is viewed in terms of females giving rise to new females. **Table 53.1** is a life table built in this way for female Belding's ground squirrels (*Urocitellus beldingi*) from a population located in the Sierra Nevada mountains of California. Next, we'll take a closer look at some of the data presented in a life table.

Survivorship Curves

The survival rate data in a life table can be represented graphically as a **survivorship curve**, a plot of the proportion or numbers in a cohort still alive at each age. As an example, let's use the data for female Belding's ground squirrels in Table 53.1 to draw a survivorship curve. Often, a survivorship curve begins with a cohort of a convenient size—say, 1,000 individuals. To obtain the other points in the curve for the ground squirrel population, we multiply the proportion alive at the start of each year (the third column of Table 53.1) by 1,000 (the hypothetical beginning cohort). The result is the number alive at the start of each year. Plotting these numbers versus age for female Belding's ground squirrels

Table 53.1 Life Table for Female Belding's Ground Squirrels (Tioga Pass, in the Sierra Nevada Mountains of California)

Age (years)	Number Alive at Start of Year	Proportion Alive at Start of Year*	Death Rate†	Average Number of Female Offspring per Female
0–1	653	1.000	0.614	0.00
1–2	252	0.386	0.496	1.07
2–3	127	0.197	0.472	1.87
3–4	67	0.106	0.478	2.21
4–5	35	0.054	0.457	2.59
5–6	19	0.029	0.526	2.08
6–7	9	0.014	0.444	1.70
7–8	5	0.008	0.200	1.93
8–9	4	0.006	0.750	1.93
9–10	1	0.002	1.00	1.58

▲ Researchers working with a Belding's ground squirrel

Data from P. W. Sherman and M. L. Morton, Demography of Belding's ground squirrel, *Ecology* 65:1617–1628 (1984).

*Indicates the proportion of the original cohort of 653 individuals that are still alive at the start of a time interval.

†The death rate is the proportion of individuals alive at the start of a time interval that die during that time interval.

Mastering Biology Table Walkthrough